

# Timothy Do

An A.I. Machine Vision Enthusiast & DSP Techie to Make the World A Better Place



## EDUCATION

### University of California, Irvine — B.S. Electrical Engineering

September 2019 - June 2023 (Expected)

- 4th Year Undergraduate, Accelerated Status
- Overall GPA: 3.90, Major GPA: 3.91
- **Specialization:** Digital Signal Processing (DSP)
- **Research Interests:** DSP with AI, Image Enhancements such as Colorization & Motion Interpolation with Neural Networks
- **Relevant Coursework:**
  - Electronics (EECS 170A/B/C), Digital Signal Processing (EECS 152A/B), Digital Image Processing (EECS 203A)
- **Major Coursework** for Fall 2022: EECS 159A (Senior Design Project), EECS 160A/LA (Control Systems), ENGR 190W (Engineering Writing)
- **Clubs and Societies:** ICVL, IEEE-HKN, IPC, TBP
- **Awards:** Dean's Honor List (GPA 3.5+) for all nine quarters at UCI

## EXPERIENCE

### Western Digital, San Jose, CA - Industrial IOT Intern

June 2022 - September 2022

- **Skills:** MySQL, Python, Raspberry Pi, Soldering, Linux, Lean Six Sigma
- Certified as a Six Sigma Yellow Belt for learning Six Sigma foundations.
- Developed Presentation Skills in a Corporate Environment.
- Learned Industrial Tool Sets for Circuit Design and Embedded System Programming.

### UC Irvine, Irvine, CA - Student Researcher

January 2022 - Present

- Assisted in research under the direction of Professor Glenn Healey on the topic of Spectral Image Filtering for Emissions on Wildfires
- Digitized Imaging Filters into CSVs and compared them in MATLAB against certain chemical emissions.
- Generated altitude maps using QGIS & SRTM 1 Arc-Second Global

### IPC Club @ UCI, Irvine, CA - Vice President

November 2021 - Present

- Reserved club meeting rooms and sent weekly emails to club members regarding club events
- Programmed the club website and managed social media (i.e. moderator of club discord server)
- Led a Two-Part AM Radio Workshop where I had to design the circuit for the best amplifier gain in addition to creating easy-to-read slides & ordering parts from Amazon/Dollar Tree.

### UC Irvine D.C.E., Irvine, CA - I.T. General Assistant

July 2021 - June 2022

- Helped Set-Up Desktops, Docks, Printers, Webcams, and Computers for over 300+ Employees in UCI's Division of Continuing Education
- Worked with Network Administrators by managing IPs, Subnet Masks, and DNS servers of DCE to access local networking services
- Monitored and updated a real-time web server (DCE-OLLI) by rebuilding the frontend/backend using react.js.

## CONTACT

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[LinkedIn](#) (/in/do-timothy)

[Github](#) (dotimothy)

<https://timothydo.me>

## SKILLS

### Computers:

Assembled/Debugged Desktop Hardware and OS installations.

### Electronics:

Soldered Circuits and Designed Components on Oscilloscopes, Power Supplies, Function Generator and Multimeter.

### Embedded System Programming:

Programmed with C, Java, and Python libraries by creating IoT projects on Arduino/Raspberry Pis

### Image Processing & Machine Learning:

Programmed GPUs to do Image Processing in MATLAB & Tensorflow AI

### Web Socket Programming:

Created Websites with HTML, CSS, and Javascript in addition to TCP socket programming using Python.

## PROJECTS

**EECS195-Colorization:** Colorizing Images using GANs & CNNs.

**Languages:** MATLAB, Python

**PiFmMorse:** A Morse Code FM Radio Transmitter for the Raspberry Pi

**Languages:** GPIO Python

**TTLZNetChess:** A C-Based Chess Engine with Minimax AI & TCP Networking

**Languages:** C, Makefile

**PhotoLabServer:** A Flask Based Server for Advanced Image Processing

## LANGUAGES

**English** (Fluent), **Vietnamese** (Fluent), & **Spanish** (Proficient)